

DATA ANALYSIS & BUSINESS INTELLIGENCE PROJECT

BMW Sales Dashboard

From messy raw data to a clean, interactive Power BI dashboard — cleaned with Pandas and modeled in Power BI.

PROJECT TEAM

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The Project, in Short

Turning a raw sales export into a dashboard people can actually use.

This project looks at five years (2019–2023) of global BMW sales — **5,778 raw records** across 23 countries — and turns them into a clean, interactive dashboard that answers real business questions: which models sell, which markets lead, and where the growth opportunities are.

- Python 3
- Pandas
- Power BI Desktop
- DAX
- Star Schema

 **Stage 1 — Pandas**

The raw CSV was checked for missing values, wrong data types, and duplicates, then cleaned with a short reusable script — before it ever touched Power BI.

 **Stage 2 — Power BI**

The clean data was modeled as a star schema and turned into an executive dashboard with drill-down pages, using DAX for growth and time comparisons.

The Dataset

What we started with



Each row is one sale: **Date**, **Model**, **Revenue**, **Quantity Sold**, **Region**, **Country** and **Channel** (Wholesale, Dealership, or Online) — covering everything from the BMW 1 Series to the electric iX.

What Was Wrong, and How Pandas Fixed It

778 rows removed, 1 new column added

- 1 **778 completely empty rows** (13.5% of the file) — dropped with `dropna(how='all')`.
- 2 **Dates stored as plain text** — converted to real dates so we could analyze trends and growth over time.
- 3 **Year and Quantity Sold stored as decimals** (e.g. 2019.0) — fixed to whole numbers.
- 4 **No price-per-unit column** existed — we added **Revenue Per Unit** for model comparisons.
- 5 **Messy text formatting** in Model, Region, Country, and Channel — trimmed and standardized.

No duplicate transactions, negative values, or spelling issues were found underneath — once the blank rows were gone, the data was clean and consistent.



The Cleaning Script

A short, reusable Pandas script — the core of the whole pipeline.

```
import pandas as pd

df = pd.read_csv('BMW_Sales_Data.csv')

df = df.dropna(how='all')           # drop fully empty rows
df = df.drop_duplicates()         # drop duplicate rows
df = df.dropna(subset=['Date', 'Model', 'Revenue',
                      'Quantity Sold', 'Region', 'Country', 'Channel'])

df['Date'] = pd.to_datetime(df['Date'], format='%d/%m/%Y')
df['Year'] = df['Date'].dt.year
df['Revenue'] = df['Revenue'].astype(float)
df['Quantity Sold'] = df['Quantity Sold'].astype(int)
df['Revenue Per Unit'] = (df['Revenue'] / df['Quantity Sold']).round(2)

for col in ['Model', 'Region', 'Country', 'Channel']:
    df[col] = df[col].str.strip()

df = df.sort_values('Date').reset_index(drop=True)
df.to_csv('BMW_Sales_Data_Cleaned.csv', index=False)
```

Seven clear steps: drop the junk, fix the types, add one useful column, tidy the text, and save. That's the entire cleaning pipeline.

Questions & KPIs

What we set out to answer

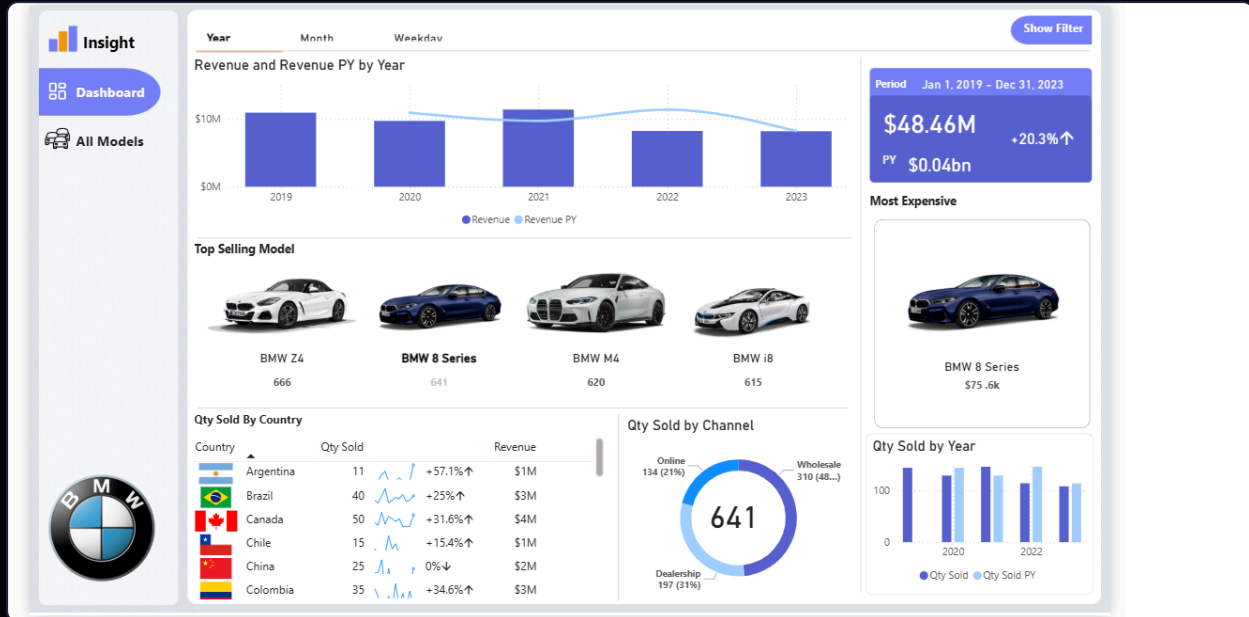
- **Q1.** How is revenue trending year over year — growing, shrinking, or flat?
- **Q2.** Which models lead in revenue, and do they lead in units too?
- **Q3.** Which sales channel (Wholesale / Dealership / Online) drives the most revenue?
- **Q4.** Which regions and countries drive the most sales?
- **Q5.** Is there seasonality — do some months outperform others?
- **Q6.** How much does average price vary by model?

KPI	WHY IT MATTERS
Total Revenue	Headline measure of overall performance
Total Units Sold	Demand, independent of price
Revenue Growth %	Shows momentum — speeding up or slowing down
Average Price / Unit	Signals brand and model positioning
Revenue by Channel / Region	Guides where to invest next

The Dashboard

The final Power BI report, built on the cleaned data.

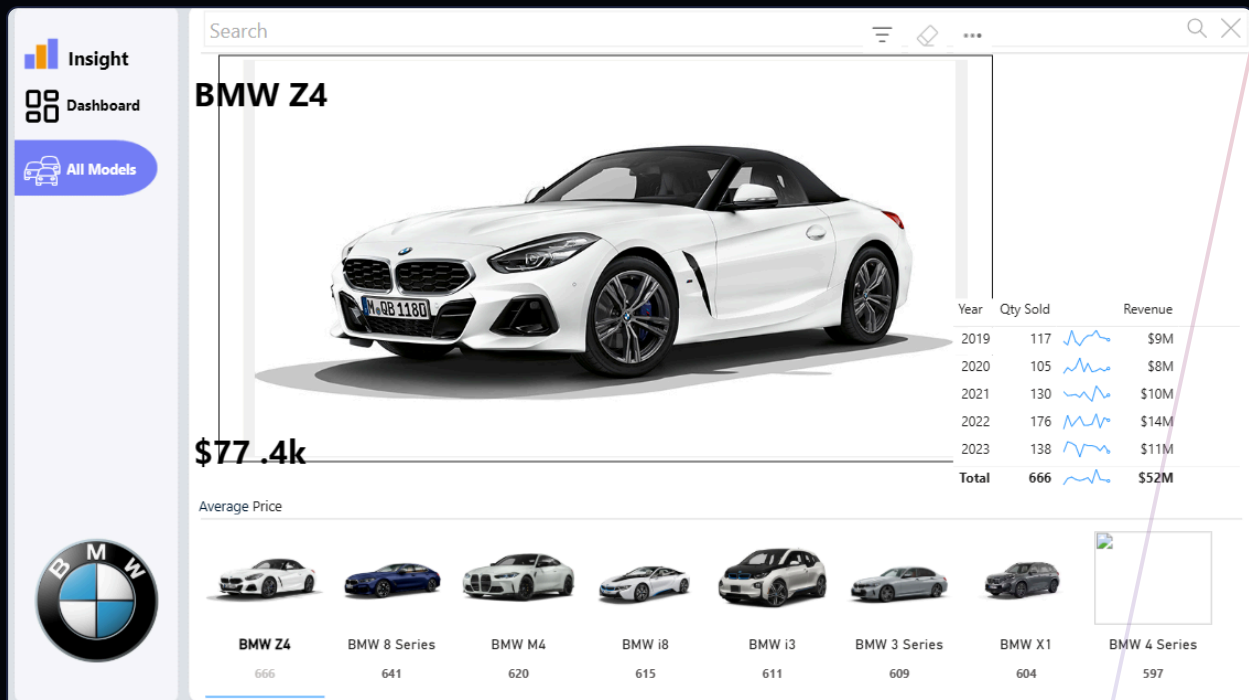
Executive Dashboard



Global revenue trend, top selling models, and channel breakdown — all in one view.

This page answers the big-picture questions at a glance: **\$1.13bn cumulative revenue** (+24.7% vs. the prior period), **BMW Z4** as the top-selling model at 666 units, and a channel split dominated by Wholesale (45%) over Dealership (33%) and Online (22%).

All Models — Drill Down



Click into any model to see its year-by-year performance — shown here for the BMW Z4.

This page lets anyone click a model and see its full history — year, quantity, and revenue — with a trend line for each year. It's how we caught that some models (like the 8 Series) peaked in 2021 and have been declining since.

Key Findings

What the clean data actually tells us.

\$376.1M

TOTAL REVENUE

15,002

UNITS SOLD

\$34,187

AVG. PRICE / UNIT

-2.0%

2023 YOY REVENUE

 **Revenue is flat, not growing**

Sales peaked in 2021, then softened two years straight (~1.0% in 2022, -2.0% in 2023). The brand isn't shrinking, but it's plateaued.

 **BMW Z4 leads everything**

Top model by both revenue (\$17.1M) and units (666) — followed closely by the 3 Series and X4.

 **Wholesale dominates revenue**

44% of revenue comes from Wholesale, vs. only 22% from Online — a heavy tilt toward bulk deals over direct sales.

 **Africa leads, Europe lags**

Africa brings in the most revenue (21.9%), while Europe — BMW's home market — trails every region (18.7%).

 **August is the peak month**

Revenue peaks in August and dips in February — a modest ~15% seasonal swing worth planning around.

 **SUVs command top prices**

The BMW X3 has the highest average price (\$38,186), followed by the M8 and X4.

BMW Sales Dashboard

— 05 —

Recommendations & Wrap-Up

What we'd tell the business to do next.

01 · Grow the Online channel

It's the smallest channel (22%) but likely the cheapest to scale. A stronger online buying experience could shift volume away from Wholesale and improve margins.

02 · Double down on the Z4, review the 5 Series

The Z4 wins on both revenue and volume — give it priority. The 5 Series is the weakest performer and deserves a pricing or positioning review.

03 · Re-energize Europe

BMW's home market has the lowest regional revenue share — worth investigating competitive pressure and a refreshed local campaign.

04 · Plan around the August peak

Align inventory and launches with August/September demand, and use promotions to lift the slower February period.

05 · Push the EV line-up

Revenue has softened since the 2021 peak — a renewed push behind the i3/i4/i8/iX/iX3 range is a natural growth lever.

Conclusion

This project walked a messy BMW sales export through a complete analytics pipeline: **Pandas** for cleaning (778 blank rows removed, types fixed, one new column added), and **Power BI** for modeling and visualization. The result is a dashboard that turns 6 business questions into answers anyone can explore in seconds.

The story the data tells is simple: BMW's global sales are strong overall (\$376M across five years) but have **plateaued since 2021**, revenue leans heavily on **Wholesale** over Online, and performance varies clearly by model and region — with real, actionable opportunities in each of those gaps.

Deliverables

clean_bmw_sales.py — the Pandas cleaning script · **BMW_Sales_Data_Clean.csv** — cleaned dataset (5,000 rows) · **BMW_Dashboard.pbix** — the Power BI file · this report